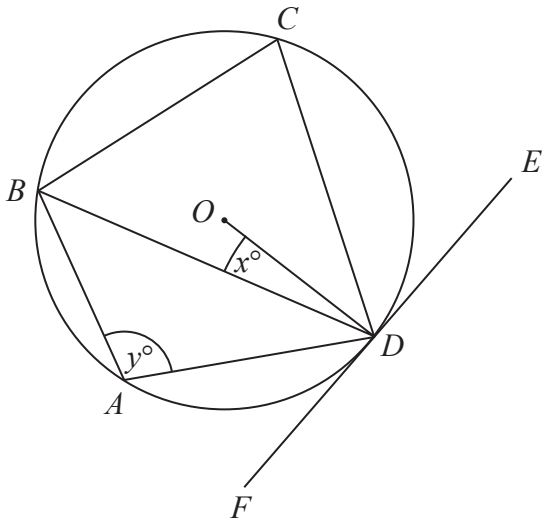
**13**




$A$, $B$, $C$ and $D$ are points on the circumference of a circle, centre $O$.
$FDE$ is a tangent to the circle.

(a) Show that $y - x = 90$
You must give a reason for each stage of your working.

Angle ODE = 90°: a tangent is perpendicular to the radius.

**1 mark**
Identify a missing angle

Angle BDE = y° by the alternate segment theorem.

But angle BDE = angle BDO + angle ODE = x° + 90°.

So y = x + 90, hence y – x = 90.

**1 mark**
Complete the angle argument

**1 mark**
Both circle reasons linked to the working

**(3)**

Dylan was asked to give some possible values for $x$ and $y$.

He said,

"$y$ could be 200 and $x$ could be 110, because 200 – 110 = 90"

(b) Is Dylan correct?
You must give a reason for your answer.

No. y is an interior angle of triangle BAD, so y must be less than 180°.

**1 mark**
200° cannot be an interior angle of a triangle

**(1)**

**(Total for Question 13 is 4 marks)**